INCH-POUND

A-A-59112A
18 June 2002
SUPERSEDING
A-A-59112
07 July 1997

COMMERCIAL ITEM DESCRIPTION

SPECTRUM ANALYZER
100 Hz to 1.8 GHz

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

1.1 This CID describes a solid state, internal preselection, programmable, 10 Hz resolution bandwidth, 100 Hz to 1.8 GHz spectrum analyzer with internal tracking generator. This CID is intended as a minimum requirement for the spectrum analyzer in which only those manufacturers that meet or surpass the following requirements will be considered as qualified.

2. SALIENT CHARACTERISTICS.

2.1 Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall conform to the requirements for Class 3 of MIL-PRF-28800.

2.2 Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 110/220 volts AC (±10%) at line frequencies of 45 to 440 Hz. Power consumption shall not exceed 650 VA. Transient state conditions shall conform to the requirements for Class 3 of MIL-PRF-28800.

2.3 Color. The color shall be that normally provided by the manufacturer.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/TILCC, 420 Second Street, Suite 100, Robins AFB, GA 31098-1640.

AMSC N/A FSC 6625

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2.4 Electromagnetic Interference (EMI). The equipment shall comply with the following EMI requirements specified in MIL-STD-461:

2.4.1 CE01: Narrow Band Requirements from 1 KHz to 15 KHz only.

2.4.2 CE02: Relaxed 20 dB from 15 KHz.

2.4.3 CS01: Full limits.

2.4.4 CS02: Full limits.

2.4.5 CS06: Full limits.

2.4.6 RE01: Relaxed 15 dB at the fundamental through the 10^{th} harmonic of the source power frequency, and exception 30 KHz to 50 KHz.

2.5 Temperature requirements.

2.5.1 Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of –40 degrees centigrade to +50 degrees centigrade.

2.5.2. Nonoperating temperature. The equipment shall meet its performance and accuracy requirements in a nonoperating environment of –40 degrees centigrade to +65 degrees centigrade.

2.6 Mean time between failures (MTBF). The equipment MTBF shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

2.7 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.8 Performance characteristics. Unless otherwise specified, all performance requirements shall apply after a 1 hour warm-up period in a laboratory ambient operating temperature.

2.8.1 Frequency range. The minimum frequency range of the spectrum analyzer with internal time base shall be 100 Hz to 1.8 GHz.

2.8.2 Frequency reference accuracy.

2.8.2.1 Aging rate. After 24 hours warm-up period, the aging shall be less than 1X10–9/day, and less than 2.5X10–7/year.

2.8.2.2 Temperature stability. The temperature stability shall be better than 2X10-8 for any temperature change between 0 degrees centigrade and +55 degrees centigrade.

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2.8.3 Frequency readout accuracy. The readout frequency shall be plus or minus (readout frequency X reference frequency accuracy +5% of span of RES BW plus 350 Hz).

2.8.4 Frequency counter. The spectrum analyzer shall provide a fully integrated frequency counter capability. The total accuracy shall be defined as: ±(counter frequency X reference frequency accuracy + 50 Hz + 1LSD).

2.8.5 Frequency span. The minimum span range shall be 0 Hz, 100 Hz to 1.8 GHz. The frequency span accuracy shall be better than ±5% over all spans.

2.8.6 Resolution bandwidth. The resolution bandwidth shall be 10 Hz to 2MHz (3 or 6 dB). The setting shall be front-panel selectable or automatic.

2.8.7 Residual FM. The total residual FM shall be less than 10 Hz peak to peak in 20 milliseconds.

2.8.8 Frequency response. When tuned to the specified frequency range with the attenuator set as 10 dB, the spectrum analyzer shall have the frequency response of ±1.5 dB.

2.8.9 Noise sidebands: The sideband noise shall be less than –86 dBc at 10 KHz offset from the carrier, -100 dBc at 30 KHz offset from carrier, and –110 dBc/Hz or less at 100 KHz offset from center frequency.

2.8.10 Signal input range. The minimum amplitude reference level range shall be from –90 dBm to +30 dBm with a safe input of +30 dBm in the log mode or 1 watt in the linear mode.

2.8.11 Input attenuator accuracy. The input attenuator shall have a minimum range of 0 to 70 dB. The attenuator accuracy shall not exceed 3.0 dB over 70 dB range from 100 Hz to 1.8 GHz.

2.8.12 RF input connector. The RF input connector shall be compatible with the type N connectors.

2.8.13 Input impedance. The input impedance shall be 50 ohms nominal. The voltage standing wave ratio (VSWR) shall not be greater than 3:1.

2.8.14 Input power. The spectrum analyzer shall accept alternating signals that are less than 1 watt (+30 dBm) continuous over the frequency range of 100 Hz to 1.8 GHz.

2.8.15 External reference input. The equipment shall be provided with an input to accommodate a 10 MHz external reference.

2.8.16 Output. The equipment shall be provided with an internal calibrator signal output, 1st LO output, and IF output.

2.8.17 Vertical scaling. Linear or log vertical scaling shall be selectable from the front panel. The log scale shall have a minimum of 2 selectable calibrated scales.

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2.8.17.1 2 dB/Division

2.8.17.2 10 dB/Division

2.8.18 Display accuracy. The maximum allowable cumulative error in the log display mode shall be ±4.0 dB. The maximum display accuracy of full scale display in the linear mode shall be ±5%.

2.8.19 Third order intermodulation distortion. The third order intermodulation distortion shall be –78 dBc or less with a mixer level of –30 dBm.

2.8.20 Harmonic distortion. The second harmonic distortion shall be –70 dB or less with a mixer level of –40 dBm for frequency range of 10 MHz to 1.8 GHz and –60 dBc for frequency range of 100 Hz to 10MHz.

2.8.21 Local oscillator emissions. The LO emissions reference to the input mixer shall be –80 dBm or lower with 10 dB RF attenuation.

2.8.22 Calibrator. The spectrum analyzer shall have a calibrator output to provide performance verification as determined by manufacturer.

2.8.23 IF output. The spectrum analyzer shall have an IF output, 50 ohm nominal impedance, output level is –5 dBm.

2.8.24 1st LO output. The spectrum analyzer shall provide access to the output of the respective local oscillator, greater than +5 dBm, and 50 ohm nominal impedance.

2.8.25 Sweep time. The sweep time shall be front-panel controlled for manual or automatic mode. In the manual mode the timing shall be front-panel controlled for a minimum timing range of 50 microseconds to 60 seconds.

2.8.26 Sweep time accuracy. The sweep time accuracy shall be at least ±15% for sweep time less than 30 milliseconds, ±1% for sweep time from 30 ms to 60 seconds.

2.8.27 Digital storage. Digital storage shall be provided with selectable modes that save maximum signal values and noise-average spectral displays of two signals. The digital storage function shall be capable of storing and displaying at least eight spectrums including the CRT readout measurement parameters. When a battery is required for the digital storage circuitry, the battery life shall be at least 12 months under the normal operating conditions.

2.8.28 Markers. The equipment shall provide frequency and amplitude markers, using one marker for absolute measurements and two markers for relative measurements. Marker positioning shall provide a readout of frequency and amplitude of any point, or the difference in amplitude or frequency between two points within the frequency range of the instrument.

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2.8.29 CRT readout. The CRT shall be capable of displaying the following readout.

2.8.29.1 Reference level.

2.8.29.2 Center frequency.

2.8.29.3 Vertical display mode.

2.8.29.4 Frequency span/div or frequency span.

2.8.29.5 Resolution bandwidth.

2.8.29.6 RF attenuation.

2.8.30 Tracking generator. The built-in tracking generator shall have the following characteristics:

2.8.30.1 Frequency range. The frequency range for the tracking generator shall be 300 KHz to 1.8 GHz with an accuracy of +(frequency reference accuracy X center frequency +5% X span + 300 Hz).

2.8.30.2 Minimum resolution bandwidth. The minimum resolution bandwidth shall be at least 300 Hz.

2.8.30.3 Output level. The output level for the built-in tracking generator shall be –10 dBm to 0 dBm with a resolution of 0.1 dB, level flatness of 2.0 dB, and a total absolute accuracy of ±3.25 dB.

2.8.30.4 Remote control. The spectrum analyzer shall provide an IEEE standard 488 interface function and connector. This is to enable the spectrum analyzer for remote control of all front panel functions except power switch, CRT intensity, frequency calibration, level calibration and trigger level adjustment knob.

2.8.30.5 Programmability. The spectrum analyzer shall be fully programmable by the use of IEEE standard 488, General Purpose Interface Bus (GPIB) except for CRT display positioning, CRT visual controls, manual scan, level calibration, and power.

2.8.31 Front panel controls. The following shall be front-panel controllable:

2.8.31.1 Store display.

2.8.31.2 Erase display.

2.8.31.3 Operate in non-storage display mode.

2.8.31.4 Store and display two trace, one active and one previously stored spectrum.

2.8.31.5 Perform normalization or trace subtraction.

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2.8.31.6 Peak hold.

2.8.31.7 Selective display and blanking.

2.8.31.8 Internal preselection.

2.8.31.9 Digital averaging.

2.8.31.10 Vertical display.

2.8.31.11 Linear display.

2.8.31.12 Logarithmic display.

2.8.31.13 Intensity.

2.8.32 Accessories. Each instrument shall be provided with the following.

2.8.32.1 Spare fuses.

2.8.32.2 115 VAC power cable (USA).

2.8.32.3 Operator manual.

2.8.32.4 Programmer manual.

2.8.32.5 BNC (female) to BNC (female) connector 50 ohm coax cable (length 6 feet).

3. REGULATORY REQUIREMENTS.

The offeror/contractor is encouraged to use recovered material to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.1 The equipment shall be bid sample tested within the accuracies, limits, and specifications herein.

4.2 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product is manufactured to metric dimensions exceeding the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6.2 Source of documents. Federal and Military specifications are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.3 Ordering data. Procurement documents must specify the title, number, and date of this CID, issue of DODISS to be cited in the solicitation, and preservation and packaging requirements.

Preparing Activity:
Air Force – 84

Agent Activity:
Air Force – 99

Project Number: 6625-0906

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